INDUSTRIAL INTELLIGENCE · COMMAND CENTER

What industrial ambition should we model?

Describe a product, a volume or a place. QuantarisAI models the ecosystem needed to make it economically viable, environmentally responsible and circular.

AMBITION

Manufacture 1 million cotton T-shirts annually

STARTING REGION · OPTIONAL

India

Missing context — capacity, price point, constraints — is requested during the run, not up front.

Initiate scenario →

WHAT GETS MODELLED

Product → Materials → Processes → Machines → Factory → Location → Resources → Economics → Impact → Waste → ✓ Circularity → ↗ Next opportunity

SUGGESTED MISSIONS

NEW PRODUCT · LOCATION

I want to manufacture electric motors in India.

VOLUME TARGET · DEMO SCENARIO

I want to manufacture 1 million cotton T-shirts annually.

REGIONAL DISCOVERY

What manufacturing opportunities exist in this region?

FACTORY EVOLUTION · PREVIEW

What else could this factory manufacture?

WASTE TO VALUE

Find economic uses for this factory's waste.

RECENT SCENARIOS — View all

Cotton T-shirts · 1M units / yr — ✓ COMPLETE

Recommended: Tiruppur, Tamil Nadu · 3 circular pathways

MEDIUM | 11/11 STAGES | 4 MIN AGO

Electric motors · India — ◉ RUNNING SIMULATION

Comparing 4 candidate states · waiting on copper price input

○ UNKNOWN | STAGE 6/11 | NOW

Foundry sand · reuse options — ✓ PATHWAY FOUND

Waste-to-value mission · 2 matched industrial consumers

HIGH | 5/5 STAGES | 2 DAYS AGO

Aluminium die-cast housings — ▲ FAILED · ECONOMICS

Tariff dataset unavailable. Results through Factory are preserved.

Retry economics | Open last stable stage

≈ ILLUSTRATIVE DATA · NOT LIVE

Scenarios use deterministic demonstration data. Every figure carries its evidence kind.

SCENARIO **Cotton T-shirts · 1M units / yr** STAGE Overview CONFIDENCE Medium UPDATED 4 min ago ≈ ILLUSTRATIVE DATA · NOT LIVE

Open evidence

AMBITION

Manufacture 1 million cotton T-shirts annually

✓ Recommendation ready Confidence: Medium 11 of 11 stages · run #0147 · 2 min 41 s

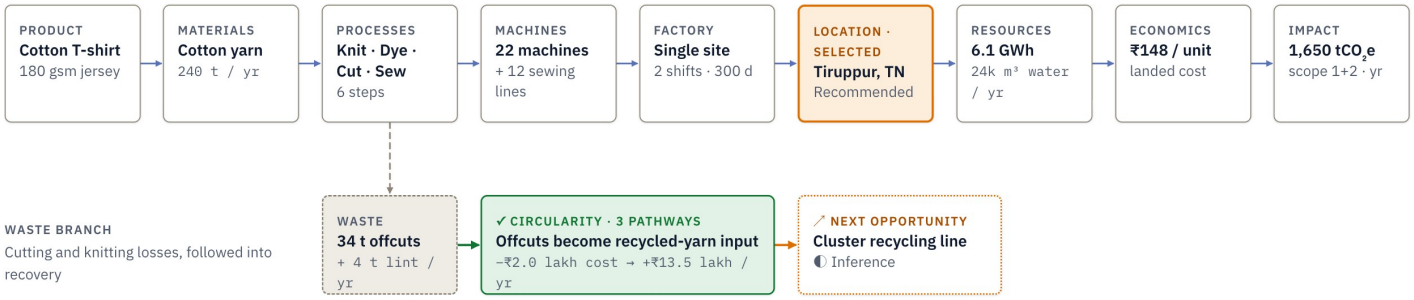
Compare locations | Review recommendation →

✓ Retrieving data 0:18 | ✓ Comparing locations 0:44 | ✓ Running simulation 1:12 | ✓ Analysing impact 1:49 | ✓ Circular pathway found 2:16 | ✓ Generating recommendation 2:35 | ✓ **Completed** 2:

Run summary

Industrial system

MATERIAL FLOW ----- WASTE BRANCH —— ✓ PROVEN VALUABLE SELECT A NODE TO INSPECT



Scenario narrative

◐ INFERENCE · SUMMARY

A 1M-unit T-shirt plant needs about 240 t of cotton yarn a year, six process steps and 336 sewing operators across two shifts. Tiruppur gives the shortest supplier distances and the lowest modelled emissions; its binding constraint is zero-liquid-discharge dyeing. Cutting leaves 34 t of clean cotton offcuts a year — enough to supply a recycled-yarn pathway instead of landfill.

◉ MOST USEFUL DECISION NOW

Confirm Tiruppur — and verify ZLD effluent capacity first

The ₹3/unit lead over Surat sits inside the uncertainty band. The effluent constraint decides.

Compare candidates

02 LOCATION	03 FACTORY	04 ECONOMICS	05 IMPACT	06 CIRCULARITY
Tiruppur	1.08M cap.	₹148 / unit	1,650 tCO2e	✓ 34 t / yr
Σ CALCULATED	Σ CALCULATED	Σ CALCULATED	∿ SIMULATION	Σ CALCULATED

→ RECOMMENDATION ✓ READY

RECOMMENDED LOCATION

Tiruppur, Tamil Nadu

- ■ FACT — Knit, dye and print job-work within 20 km
- Σ CALC — Lowest landed cost: ₹148/unit (Surat ₹151)
- ∿ SIMULATION — Lowest scope 1+2 emissions of 3 candidates
- ◐ INFERENCE — 3 recycled-yarn buyers within 80 km

Confidence: Medium
Cost lead is within uncertainty; ZLD capacity is not yet verified.

Open evidence | Compare

SCENARIO **Cotton T-shirts · 1M units / yr** STAGE Location CONFIDENCE Medium UPDATED 4 min ago ≈ ILLUSTRATIVE DATA · NOT LIVE Open evidence

LOCATION INTELLIGENCE · 3 CANDIDATES · INDIA

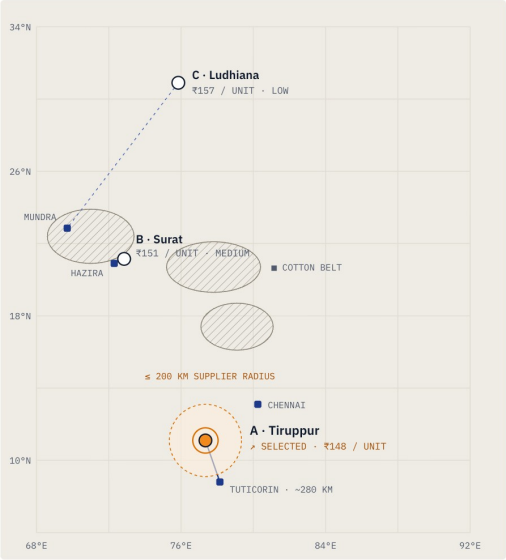
Where should this factory be?

Selecting a candidate updates the Factory, Resources, Economics and Impact stages.

Map Compare

Candidate schematic

SCHEMATIC · NOT TO SCALE



SELECTED SITE CANDIDATE PORT COTTON BELT

LIVE MAP OUT OF MVP SCOPE

Candidate comparison

No composite score — trade-offs stay visible per criterion.

CRITERION	↗ A · SELECTED Tiruppur, TN	B Surat, GJ	C Ludhiana, PB
Landed cost / unit Σ CALCULATED	₹148 range 139–161	₹151 range 141–164	₹157 range 145–171
Logistics to port Σ CALCULATED	Tuticorin ~280 km ₹9 / unit	Hazira ~25 km ₹6 / unit	Mundra ~1,150 km, rail ₹17 / unit
Cotton yarn supply ■ FACT	Spinning cluster within 100 km	Gujarat cotton belt	Punjab spinning mills
Workforce ■ FACT	Large knitwear-skilled pool	Large textile pool, synthetics-led	Knitwear-skilled, seasonal demand
Supplier proximity ■ FACT	Knit, dye, print job-work ≤ 20 km	Weaving-led; fewer knit dyers	Knit job-work ≤ 30 km
Infrastructure ■ FACT	Common effluent plants with ZLD	Common effluent plants available	Effluent capacity constrained
Environmental constraint ■ FACT	▲ ZLD required for dyeing; high groundwater stress	Discharge norms apply	▲ River-pollution scrutiny
Emissions, scope 1+2 ∿ SIMULATION	1,650 tCO_2e / yr	1,720 tCO_2e / yr	1,860 tCO_2e / yr
Confidence	Medium ZLD capacity unverified	Medium Knit supplier depth unclear	Low Sparse effluent data

Select any value to open its source, assumptions and calculation in the Evidence Drawer.

↗ ACTIVE CANDIDATE

A · Tiruppur, Tamil Nadu

✓ FACTORY ✓ RESOURCES ✓ ECONOMICS ✓ IMPACT UPDATED FOR TIRUPPUR

Why Tiruppur? Make Surat active Continue to factory →

SCENARIO **Cotton T-shirts · 1M units / yr** STAGE Factory CONFIDENCE Medium UPDATED 4 min ago ≈ ILLUSTRATIVE DATA · NOT LIVE

Open evidence

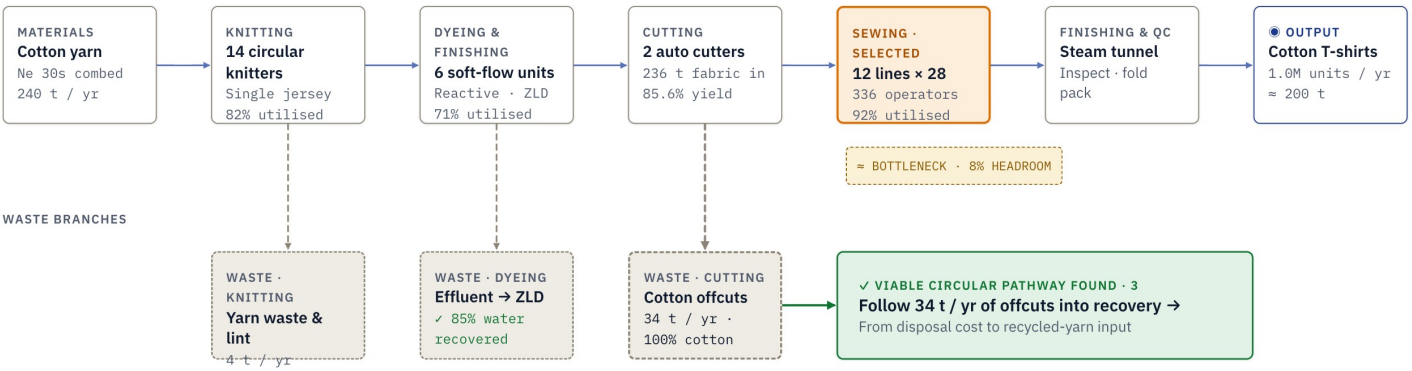
FACTORY BLUEPRINT · CONFIGURATION A

Building factory topology — Tiruppur, 1.0M units / yr

Single site · 2 shifts × 300 days · 22 machines + 12 sewing lines

MATERIAL FLOW · WASTE BRANCH · ✓ RECOVERY · ≈ ATTENTION

MATERIAL FLOW · MATERIALS → PROCESSES → MACHINES → OUTPUT



SELECTED NODE · MACHINE GROUP

Sewing — 12 lines × 28 operators

Open evidence

CAPACITY	UTILISATION	OPERATORS	MINUTES / TEE	SHIFT PATTERN
1.08M / yr	92%	336	6.2 SAM	2 × 8 h
Σ CALCULATED	Σ CALCULATED	Σ CALCULATED	◇ ASSUMPTION	◇ ASSUMPTION

≈ **Attention:** 8% headroom absorbs little demand variation. A 13th line lifts headroom to 15% for about ₹38 lakh capex. Σ CALCULATED

Capacity & constraints

KNITTING — 82% utilised — Σ CALC

DYEING — ▲ Must connect to ZLD effluent treatment — ■ FACT

DYEING — ✓ Water intake 24,000 m³ / yr, 85% recycled — Σ CALC

CUTTING — Offcut rate 14.4% of fabric — Σ CALC

SEWING — ≈ 92% utilised — bottleneck — Σ CALC

SITE — ≈ 4,200 m² built-up area — ∿ SIMULATION

SCENARIO **Cotton T-shirts · 1M units / yr** STAGE Economics & impact CONFIDENCE Medium UPDATED 4 min ago ≈ ILLUSTRATIVE DATA · NOT LIVE

Open evidence

ECONOMICS · TIRUPPUR VS ALTERNATIVES

What changes the decision?

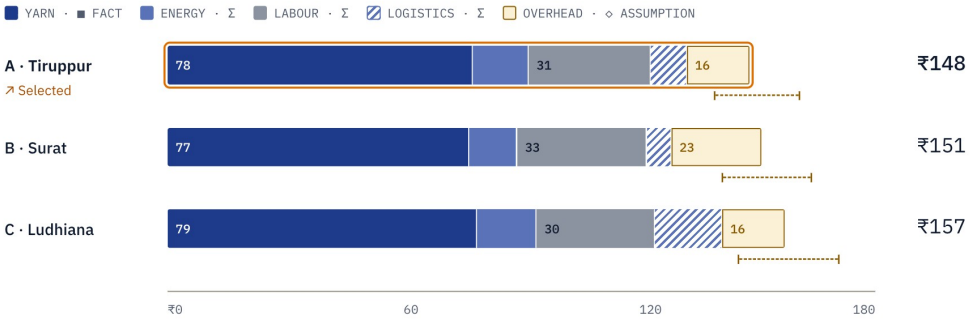
Tiruppur leads Surat by ₹3 per unit — inside the uncertainty band.

Cost alone does not separate them. The decision turns on effluent (ZLD) cost: if it rises by more than 40%, Surat becomes cheaper. Yarn price moves every site together, so it does not change the ranking.

LANDED COST	INVESTMENT	PAYBACK
₹148 / unit	₹18.4 cr	≈ 5.0 yr
Σ CALCULATED	∿ SIMULATION	◇ AT ₹185 PRICE

Landed cost per unit, by location

₹ / T-SHIRT · DASHED WHISKER = ≈ RANGE



Summary: yarn is 50–53% of cost at every site. Surat's shorter port haul is offset by higher overheads; Ludhiana carries the longest haul at ₹17 per unit. All three ranges overlap.

Sensitivity · Tiruppur

CHANGE IN ₹ / UNIT · ∿ SIMULATION

Yarn price ±15%
Moves all sites — no flip
±11.7

Effluent (ZLD) cost +40%
≈ Flips the decision to Surat
+3.2

Sewing wage +10%
Similar at all sites — no flip
+3.1

Energy tariff +20%
Surat +2.4 — no flip
+2.8

Port logistics +30%
Gap to Surat narrows to ₹2.1
+2.7

Open cost model assumptions

05 · ENVIRONMENTAL IMPACT · TIRUPPUR

Connected consequences of this configuration

Illustrative estimates. Each value is marked by label, glyph and fill pattern — calculated (solid), simulated (dotted), inferred (hatched) — never by colour alone.

ENERGY ∿ SIMULATION
6.1 GWh / yr
1.9 GWh electricity · 4.2 GWh thermal for dyeing · 6.1 kWh per tee
Open evidence

WATER Σ CALCULATED
24,000 m³ / yr
Intake · ✓ 85% of dye-house water recycled through ZLD · 24 L per tee
Open evidence

EMISSIONS ∿ SIMULATION
1,650 tCO_2e / yr
Scope 1+2 · 1.65 kg per tee · 82% from grid electricity
Open evidence

SOLID WASTE Σ CALCULATED
38 t / yr
✓ 34 t cotton offcuts recoverable · 4 t yarn waste and lint
Follow into circularity →

LOGISTICS ◐ INFERENCE
210 tCO_2e / yr
Inbound yarn and outbound to Tuticorin · road freight factors
Open evidence

CHAIN Dyeing heat & grid power → 1,650 tCO_2e → ZLD recovers 85% of water → 38 t solid waste → ✓ 34 t routable to circularity

SCENARIO **Cotton T-shirts · 1M units / yr** STAGE Circularity CONFIDENCE Medium UPDATED 4 min ago ≈ ILLUSTRATIVE DATA · NOT LIVE

Open evidence

WASTESTREAM · FROM CUTTING · 100% COTTON

Cotton offcuts → recycled-yarn input

34 t / yr Σ CALCULATED | 6 colour lots · 92% clean ■ FACT | Handled today: landfill via contractor

Compare pathways | Add pathway A to recommendation

✓ VIABLE CIRCULAR PATHWAY FOUND

Something previously treated as waste has become an economic input.

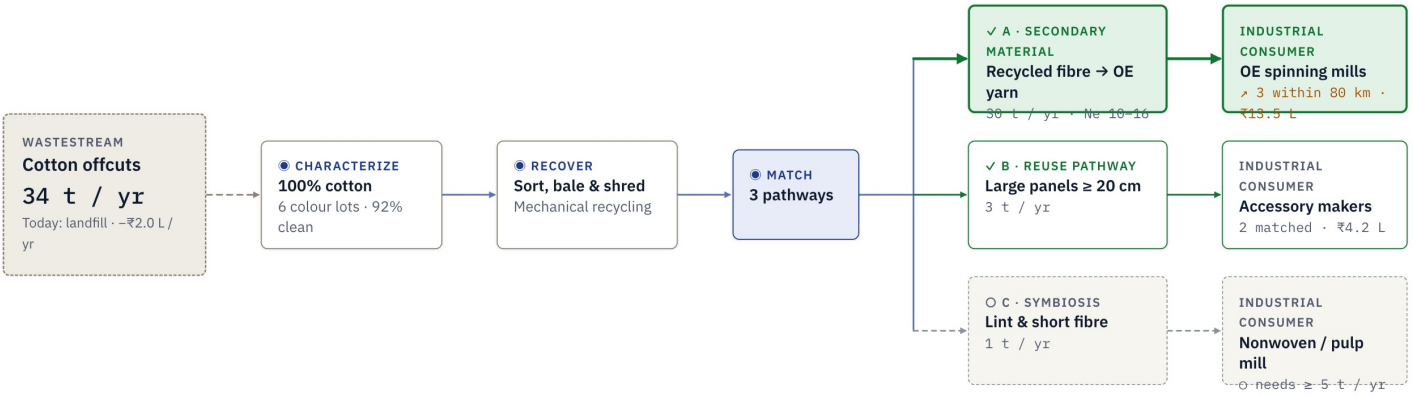
34 t a year of cutting offcuts stop costing ₹2.0 lakh in disposal and start earning ₹13.5 lakh as feedstock for recycled open-end yarn.

○ TODAY · LIABILITY
−₹2.0 L
disposal cost / yr

↗ PATHWAY A · INPUT
+₹13.5 L
secondary-material sale / yr

Recovery pathways

WASTE STATE · SYSTEM STEP · ✓ VIABLE · RECOMMENDED · ✓ VIABLE · CANDIDATE · ○ CONSTRAINED



Pathways as decisions

A and B use different fractions of the stream and can run together: ₹17.7 lakh / yr combined.

PATHWAY A — ✓ RECOMMENDED

Mechanical recycling into open-end yarn

Value	↗ ₹13.5 lakh / yr	Σ CALC
Price	₹45 / kg sorted	◇ ASSUMPTION
Avoided	✓ ≈ 45 tCO₂e / yr	∿ SIMULATION
Consumer	3 OE mills within 80 km	◐ INFERENCE

REQUIRES
- Colour-lot segregation at the cutting table
- Baling press · ≈ ₹6 lakh capex

Confidence: Medium | Evidence | Compare

PATHWAY B — ✓ VIABLE · CANDIDATE

Reuse large panels in accessories

Value	↗ ₹4.2 lakh / yr	Σ CALC
Price	₹140 / kg panels	◇ ASSUMPTION
Avoided	✓ ≈ 12 tCO₂e / yr	∿ SIMULATION
Consumer	2 accessory makers	◐ INFERENCE

REQUIRES
- 0.5 FTE sorter per shift
- Panels ≥ 20 cm kept flat and clean

Confidence: Low | Evidence | Compare

PATHWAY C — ○ CONSTRAINED

Lint and short fibre to nonwoven or pulp

Value	₹0.15 lakh / yr	Σ CALC
Volume	1 t / yr vs buyer min. 5 t	◐ INFERENCE
Avoided	Not material at this volume	∿ SIMULATION
Consumer	1 lead, unconfirmed	◐ INFERENCE

UNLOCKS IF
- Lint is pooled with 4+ nearby units
- A shared collection route exists

Confidence: Low | Evidence | Compare

↗ NEXT OPPORTUNITY ◐ INFERENCE

A cluster-scale recycled-yarn line

If about 12 comparable units in the cluster route offcuts the same way (~400 t / yr), a dedicated shredding and open-end spinning line nearby becomes viable. This factory moves from waste generator to feedstock supplier.

Model as a new ambition →

QuantarisAI Command Center Scenarios Industrial Map Factories Circularity

Cotton T-shirts · 1M units / yr Overview

Overview Location Factory Economics Impact Circularity

Manufacture 1 million cotton T-shirts annually

EVIDENCE

Why this recommendation

Dark theme ×

→ RECOMMENDATION Confidence: Medium

RUN #0147 · 4 OCT 2026, 14:32 · ≈ ILLUSTRATIVE DATA · NOT LIVE

CLAIM

Locate the 1M-unit T-shirt plant in Tiruppur, Tamil Nadu.

Medium confidence: the ₹3 cost lead over Surat is inside the uncertainty band, and ZLD effluent capacity has not been verified.

Supporting evidence

■ FACT

Established knitwear cluster: knitting, dyeing and printing job-work within 20 km.

SOURCE · illustrative cluster profile · imported 2 Oct 2026

■ FACT

▲ Dyeing units must connect to zero-liquid-discharge effluent treatment.

SOURCE · illustrative state guidance · verify current order

Σ CALCULATION

Landed cost ₹148 / unit vs Surat ₹151 and Ludhiana ₹157.

RULE · cost model v0.4 · deterministic

∿ SIMULATION

Scope 1+2 emissions 1,650 tCO_2e / yr — lowest of three candidates.

MODEL · energy & emissions v0.3 · grid factor assumed

◐ INFERENCE

Three open-end spinning mills within 80 km could take the offcuts (pathway A).

BASIS · supplier directory match · unverified

Assumptions & constraints

◇ 2 shifts × 300 days ◇ Yarn ₹325 / kg, Ne 30s

◇ Ex-factory price ₹185 — your input

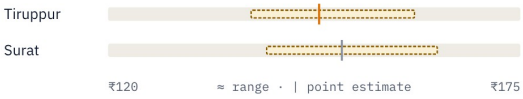
◇ Grid 0.71 $kgCO_2e$ / kWh

▲ Constraint: ZLD connection required

Calculation context

```
landed_cost = yarn 78 + energy 14 + labour 31
            + logistics 9 + overhead 16
            = ₹148 / unit
same inputs → same output · cost model v0.4
```

Uncertainty



Cost alone does not separate the two. Tiruppur leads on supplier proximity, simulated emissions and circularity. The decision flips if effluent cost rises by more than 40%.

NEXT VERIFICATION ACTION

Confirm ZLD effluent capacity and tariff with a common effluent treatment plant in Tiruppur.



Create verification task | Copy evidence link

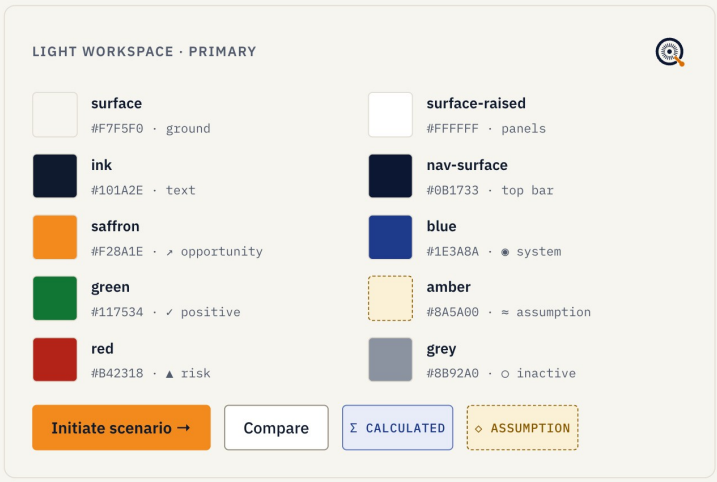
Evidence shows sources, inputs, rules and models — never private model reasoning.

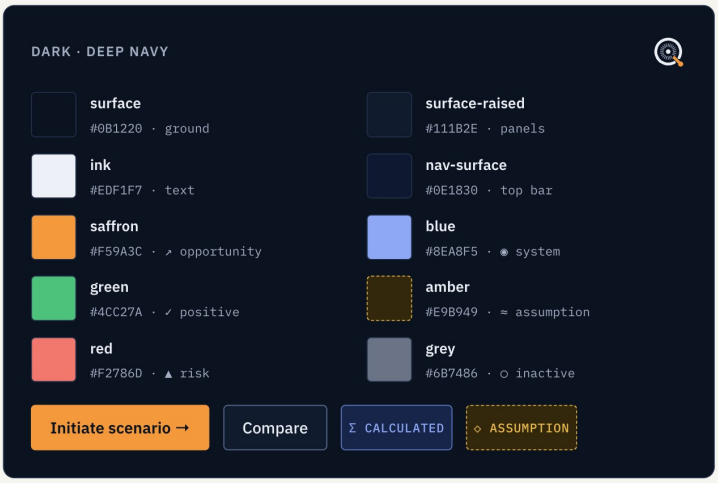
QUANTARISAI DESIGN SYSTEM · APPLIED TO VERTICAL SLICE 1

Industrial intelligence, born in Bharat

Saffron marks opportunity and the one primary action per region. Ashoka blue carries system and intelligence. India green is earned — it appears only when waste has become value. Everything else is a warm neutral workspace.

Both themes, one token set





Every screen in this canvas has a Theme tweak and a working "Dark theme / Light theme" switch in its top bar. Text on saffron is always navy (on-saffron), never white.

Type

DISPLAY · ARCHIVO 48/600	Industrial Intelligence
METRIC-XL · ARCHIVO 40/600	1,000,000
TITLE-1 · ARCHIVO 28/600	Cotton T-shirts · Tiruppur cluster
TITLE-2 · PLEX SANS 20/600	Location Intelligence
BODY · PLEX SANS 15/400	Cutting waste can become recycled yarn within 80 km.
LABEL · PLEX SANS 12/600	CIRCULARITY
METRIC · PLEX MONO 20/500	42.6 tCO_2e · ₹148 · 34 t / yr

Evidence kinds

■ FACT · ◇ ASSUMPTION · Σ CALCULATION · ◐ INFERENCE · ∿ SIMULATION · → RECOMMENDATION

Economics labels map onto these: Known = ■, Assumption = ◇, Calculated = Σ, Modelled = ∿, Inferred = ◐. Glyph, border style and word always travel together.

Status & confidence

↗ OPPORTUNITY · ◉ SYSTEM · RUNNING · ✓ COMPLETE · ✓ POSITIVE · ≈ ASSUMPTION · ▲ RISK · ○ INACTIVE

High · Medium · Low · Unknown

Flow lines — systems, not cards

- Material flow, left → right
- Waste branch, downward
- ✓ Branch proven valuable
- ○ Constrained pathway
- ↗ Selected / opportunity path
- Inferred next opportunity

Required states

LOADING · STAGE SKELETON

◉ Comparing locations · 3 candidates

○ EMPTY

No waste streams identified yet

Circularity needs a factory configuration. Run the Factory stage to identify waste streams.

Go to Factory

▲ ERROR

Economics failed: tariff dataset unavailable

Results through Factory are preserved.

Retry | Return to Factory

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Foundry sand · reuse options — ✓ PATHWAY FOUND
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HIGH · 5/5 STAGES · 2 DAYS AGO

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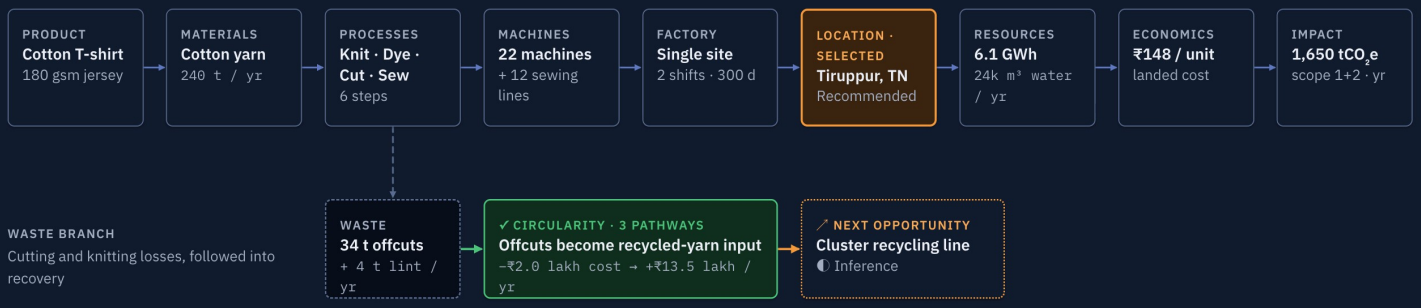
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Open evidence | Compare

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34 t a year of cutting offcuts stop costing ₹2.0 lakh in disposal and start earning ₹13.5 lakh as feedstock for recycled open-end yarn.

○ TODAY · LIABILITY

−₹2.0 L

disposal cost / yr

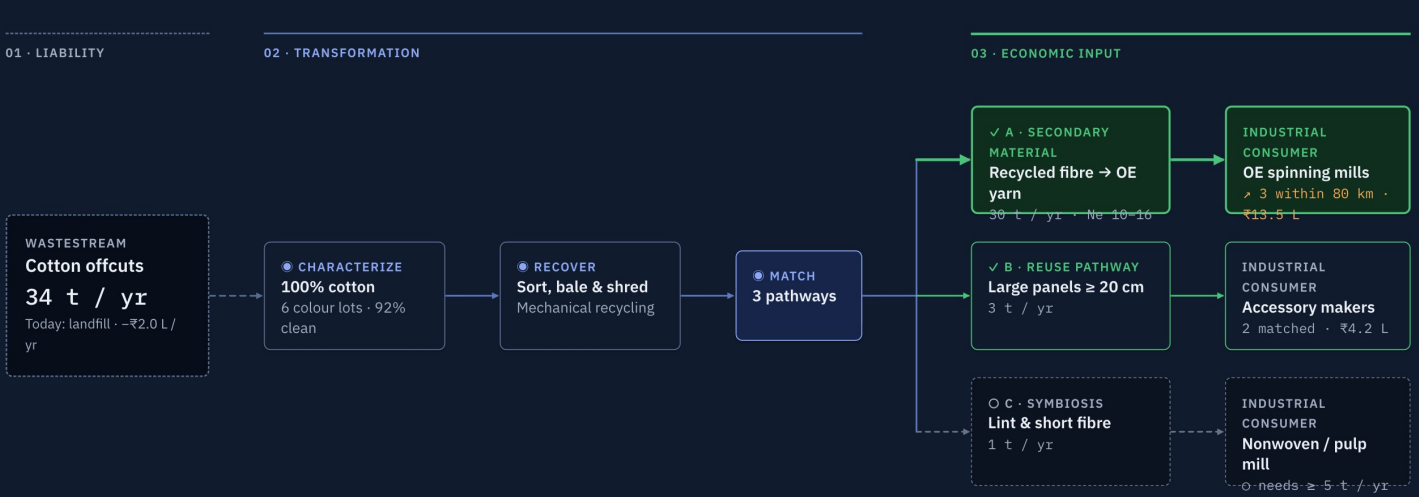
↗ PATHWAY A · INPUT

+₹13.5 L

secondary-material sale / yr

Recovery pathways

WASTE STATE · SYSTEM STEP · ✓ VIABLE · RECOMMENDED · ✓ VIABLE · CANDIDATE · ○ CONSTRAINED



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REQUIRES

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- Baling press · ≈ ₹6 lakh capex

Confidence: Medium | Evidence | Compare

PATHWAY B

✓ VIABLE · CANDIDATE

Reuse large panels in accessories

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Avoided	✓ ≈ 12 tCO₂e / yr	∿ SIMULATION
Consumer	2 accessory makers	◐ INFERENCE

REQUIRES

- 0.5 FTE sorter per shift
- Panels ≥ 20 cm kept flat and clean

Confidence: Low | Evidence | Compare

PATHWAY C

○ CONSTRAINED

Lint and short fibre to nonwoven or pulp

Value	₹0.15 lakh / yr	Σ CALC
Volume	1 t / yr vs buyer min. 5 t	◐ INFERENCE
Avoided	Not material at this volume	∿ SIMULATION
Consumer	1 lead, unconfirmed	◐ INFERENCE

UNLOCKS IF

- Lint is pooled with 4+ nearby units
- A shared collection route exists

Confidence: Low | Evidence | Compare

↗ NEXT OPPORTUNITY ◐ INFERENCE

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If about 12 comparable units in the cluster route offcuts the same way (~400 t / yr), a dedicated shredding and open-end spinning line nearby becomes viable. This factory moves from waste generator to feedstock supplier.

Model as a new ambition →

QuantarisAI Command Center Scenarios Industrial Map Factories Circularity

Cotton T-shirts · 1M units / yr Overview

Overview Location Factory Economics Impact Circularity

Manufacture 1 million cotton T-shirts annually

EVIDENCE

Why this recommendation

Light theme ×

→ RECOMMENDATION Confidence: Medium

RUN #0147 · 4 OCT 2026, 14:32 · ≈ ILLUSTRATIVE DATA · NOT LIVE

CLAIM

Locate the 1M-unit T-shirt plant in Tiruppur, Tamil Nadu.

Medium confidence: the ₹3 cost lead over Surat is inside the uncertainty band, and ZLD effluent capacity has not been verified.

Supporting evidence

■ FACT

Established knitwear cluster: knitting, dyeing and printing job-work within 20 km.

SOURCE · illustrative cluster profile · imported 2 Oct 2026

■ FACT

▲ Dyeing units must connect to zero-liquid-discharge effluent treatment.

SOURCE · illustrative state guidance · verify current order

Σ CALCULATION

Landed cost ₹148 / unit vs Surat ₹151 and Ludhiana ₹157.

RULE · cost model v0.4 · deterministic

∿ SIMULATION

Scope 1+2 emissions 1,650 tCO_2e / yr — lowest of three candidates.

MODEL · energy & emissions v0.3 · grid factor assumed

◐ INFERENCE

Three open-end spinning mills within 80 km could take the offcuts (pathway A).

BASIS · supplier directory match · unverified

Assumptions & constraints

◇ 2 shifts × 300 days ◇ Yarn ₹325 / kg, Ne 30s

◇ Ex-factory price ₹185 — your input

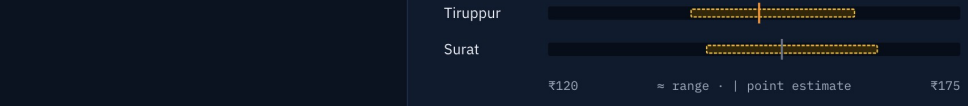
◇ Grid 0.71 $kgCO_2e$ / kWh

▲ Constraint: ZLD connection required

Calculation context

```
landed_cost = yarn 78 + energy 14 + labour 31
            + logistics 9 + overhead 16
            = ₹148 / unit
same inputs → same output · cost model v0.4
```

Uncertainty



Cost alone does not separate the two. Tiruppur leads on supplier proximity, simulated emissions and circularity. The decision flips if effluent cost rises by more than 40%.

NEXT VERIFICATION ACTION

Confirm ZLD effluent capacity and tariff with a common effluent treatment plant in Tiruppur.

Create verification task

Copy evidence link

Evidence shows sources, inputs, rules and models — never private model reasoning.